1. $\varphi(23)=22$

若 $x^2\equiv a \bmod 23$ 有解

则 a 是模 23 的二次剩余

全体二次剩余为 $1, 2^2 \cdots (\frac{23-1}{2})^2 \bmod 23$

即 1, 4, 9, 16, 2, 13, 3, 18, 12, 8, 6

非二次剩余为 5, 7, 10, 11, 14, 15, 17, 19, 20, 21, 22

2. ① $(\frac{2}{31}) = 2^{\frac{31-1}{2}} \bmod 31$

$= 2^{15} \bmod 31$

$= (2^5)^3 \bmod 31$

$= 1$

∴ 有解，解的个数为 2

② $(\frac{3}{31})(\frac{31}{3}) = (-1)^{\frac{2\times 30}{4}} = -1$

$(\frac{31}{3}) = (\frac{1}{3}) = 1$

∴ $(\frac{3}{31}) = -1$

∴ 解的个数为 0

③ $30 = 2\times 3\times 5$

∴ 有 $\begin{cases} x^2\equiv 19 \pmod 2 \\ x^2\equiv 19 \pmod 3 \\ x^2\equiv 19 \pmod 5 \end{cases}$

对 $x^2\equiv 19\equiv 1 \pmod 2$ $x\equiv 1 \pmod 2$ 解为 1 个

对 $x^2\equiv 19\equiv 1 \pmod 3$

$(\frac{1}{3}) = 1 \Rightarrow$ 解为 2 个

对 $x^2\equiv 19\equiv 4 \pmod 5$

$(\frac{4}{5}) = 4^{\frac{5-1}{2}} \bmod 5$

$= 1 \Rightarrow$ 解为 2 个

$1\times 2\times 2 = 4$ (个)

3. $7\equiv 7 \pmod{19}$

$14\equiv 14 \pmod{19}$

$21\equiv 2 \pmod{19}$

$28\equiv 9 \pmod{19}$

$35\equiv 16 \pmod{19}$

$42\equiv 4 \pmod{19}$

$49\equiv 11 \pmod{19}$

$56\equiv 18 \pmod{19}$

$63\equiv 6 \pmod{19}$

有 4 个数大于 $\frac{19}{2}$

∴ $(\frac{7}{19}) = (-1)^4 = 1$

4. $(\frac{-2}{p}) = (\frac{-1}{p})(\frac{2}{p})$

$= (-1)^{\frac{p-1}{2}} \cdot (-1)^{\frac{p^2-1}{8}}$

$= (-1)^{\frac{p^2-1+4p-4}{8}}$

$= (-1)^{\frac{p^2+4p-5}{8}}$

$p^2+4p-5 = (p+5)(p-1)$

∵ $8 \mid (p+5)(p-1)$

∴ p 为奇数，考虑二次剩余

则 $16 \mid (p+5)(p-1)$

则 $p\equiv 1 \bmod 8$ 或 $3 \bmod 8$

非二次剩余 $p\equiv 5$ 或 $7 \bmod 8$

5. ① $(\frac{17}{99}) = (\frac{17}{9})\cdot(\frac{17}{11})$

$= (\frac{8}{9})\cdot(\frac{6}{11})$

$= (\frac{-1}{9})\cdot(\frac{2}{11})\cdot(\frac{3}{11})$

$= (-1)^{\frac{9-1}{2}}\cdot(-1)^{\frac{11^2-1}{2}}\cdot(3^{\frac{11-1}{2}} \bmod 11)$

$= 1\times 1\times 3^5 \bmod 11$

$= 1$

② $650 = 2\times 5\times 61$

$987 = 3\times 7\times 47$

原式 $= (\frac{2}{3})(\frac{2}{7})(\frac{2}{47})(\frac{5}{3})(\frac{5}{7})(\frac{5}{47})(\frac{61}{3})(\frac{61}{7})(\frac{61}{47})$

$= -1$

6. $p = 2^n+1$ 为素数

$\varphi(p) = 2^n$. 3 模 p 的阶整除 $\varphi(p)$

∴ $r = 2^k$, $0\le k\le n$

若 $k<n$，由费马小定理 $3^{p-1} = 3^{2^n}\equiv 1 \bmod p$

则 $(3^{2^{k-1}})^2 \equiv 1 \pmod p$

$(3^{2^{k-1}}-1)(3^{2^{k-1}}+1)\equiv 0 \bmod p$

对于 $p = 2^{n+1} > 3$

若 $3^{2^{k-1}}\equiv 1 \bmod p$ 与 $r = 2^k$ 是 3 模 p 的阶矛盾

若 $3^{2^{k-1}}\equiv -1 \bmod p$，则 $3^{2^k}\equiv 1 \bmod p$

当 $k<n \Rightarrow$ 矛盾

∴ $r = 2^n = p-1$

∴ 3 为一个原根